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Carbon Taxes: An Explainer

Written by: Thomas Morton

Thomas

Morton

The Earth's steadily warming climate is threatening to seriously affect our way of life. If humanity is unable to rein in carbon emissions, climate change will make many places too hot for humans to live in, severely damage agricultural productivity, exhaust some freshwater supplies, and flood low-lying coastal areas. Even in the face of other

serious issues, climate change is still the biggest threat to our way of life and it is not a crisis that can be solved overnight.

Solving this issue, or at least minimizing the harm, is fundamentally an economic problem. While climate scientists build models to predict the impacts of carbon emissions, it is ultimately a matter of economic policy to work out how to reach an acceptable level of emissions while minimizing the sacrifices that will be needed in order to reach these goals. The consequences of climate change are dire, but a sudden forced decarbonization would also have unacceptably high human consequences. A balanced approach is needed – one that steadily lowers carbon intensity by making low-carbon technologies the most economical option for people and businesses.

Putting a price on carbon emissions will nudge consumers and firms into making incremental changes that reduce their carbon footprint. When I put a litre of gasoline in my car, I am paying the gas station a price that covers the costs of extracting crude oil, refining it into gasoline, and transporting it, along with the operating costs of the gas station, taxes which partially cover the costs of the road network, and a profit margin for all the companies involved in supplying the gasoline. There is an extra cost that I am not paying: the long-term cost of putting that additional carbon into the atmosphere. That extra cost is an externality, and the fact that I am not paying for it means that I am driving my car more often than I should be. If gas were substantially more expensive, I would take public transit or bike more often, and it would make an electric vehicle a more attractive option the next time I buy a car.

Changing my own behaviour will have a very small impact on cutting down on carbon emissions, but these behavioural changes extended across the economy would lead to big changes in the long run. As more people choose to live without a car, or buy electric vehicles, much less gasoline would be burned. Logistics companies would be incentivized to ship product by rail where possible, and to invest in electric vehicles.

Commercial building operators would be further incentivized to decarbonize their heating systems, and heavy industrial operations would find lower-carbon production processes more economical. [1]

While the carbon tax might be an effective way to lower carbon emissions, it would also have a serious effect on the standard of living. For many, it would be a serious hardship to face large jumps in the price of fossil fuels. Many of the uses of fossil fuels are not luxuries, and the alternatives are still costly in comparison. This would lead to even higher levels of inequality. [2] The best solution to this is counterintuitive: to return the tax revenues to the taxpayers at a flat rate. This way, the incentives to reduce consumption of fossil fuels and adopt renewable technologies still exist, but if the consumer is unable to make any immediate changes, they are still able to afford to buy as much fuel as they were before the tax. Most people would find a way to reduce their carbon emissions and spend the rebate on other things, especially once furnaces and cars need replacing.

Decarbonizing heavy industry is another special challenge for climate policy. Many carbon-intensive sectors, like steel, aluminum, and glass, are easy to ship long distances. These industries might respond to an increase in energy costs by producing elsewhere and importing the product into Canada. [3] This would be counterproductive – Canadian industry would take a hit, and the carbon emissions that are baked into our use of these materials would not change.

The most obvious answer to this problem is to apply a border adjustment, where imported goods would face tariffs that account for the carbon that would have been emitted when the product was created. This would level the playing field between domestic producers facing a carbon tax and foreign companies producing the same goods in a country without a carbon tax. This levelling is the critical part: international trade rules do not allow a country to tax imports in order to favour its own industries, but they do allow a country to apply a border adjustment that subjects imports to the same rules that

domestic producers follow. The federal government has signalled that it is looking to add a border adjustment to the existing carbon tax. This is a welcome improvement to the current policy, where emissions prices are unequal: they are set according to the final product in order to incentivize producers to reduce emissions without harming competitiveness.

Works Cited

[1] Ralph Martin, Laure de Preux, and Ulrich Wagner: *The impact of a carbon tax on manufacturing: Evidence from microdata*. The Journal of Public Economics, September 2014.

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[3] Julie Reinaud: *Trade, Competitiveness and Carbon Leakage: Challenges and Opportunities*. Chatham House Policy Institute, January 2009.

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